

Ablation Study of Recent Architectural Modifications in BERT

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Abstract

BERT(Bidirectional Encoder Representations from Transformers) has become one of the most widely used encoder-based language models due to its strong performance on various natural language processing tasks. In this study, we investigate the effectiveness of several recent architectural modifications when applied to a BERT backbone. Specifically, we evaluate SwiGLU, Rotary Positional Encoding (RoPE), and Mixture-of-Experts (MoE) under identical training conditions through an ablation study. Experimental results on Korean Wikipedia pre-training data show that all variants converge stably during training. Among the evaluated methods, SwiGLU and MoE provide modest improvements over the baseline BERT model, while RoPE shows limited benefits in the current setting. These findings suggest that activation functions and feed-forward network designs have a greater impact on BERT pre-training performance than positional encoding methods in small-scale models.

1 Introduction

Large language models aim to understand contextual relationships between tokens and sentences through large-scale pre-training. Among Transformer-based architectures, encoder-only models such as BERT and decoder-only models such as GPT have demonstrated remarkable performance across various natural language processing tasks.

BERT is pre-trained using two self-supervised objectives: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). By leveraging deep

bidirectional representations, BERT achieved strong performance on a wide range of downstream tasks and became one of the foundational architectures in modern NLP research.

In this study, we investigate the effectiveness of several recently proposed architectural techniques when applied to a BERT-based backbone. Specifically, we conduct a series of ablation experiments by independently modifying core components of the original BERT architecture while keeping all other training settings fixed.

First, we replace the original GELU activation function with SwiGLU to evaluate the impact of gated activation mechanisms. Second, we substitute the learned positional embedding with Rotary Positional Encoding (RoPE) to analyze the effect of relative positional representations. Finally, we replace the standard Feed-Forward Network (FFN) with a Mixture-of-Experts (MoE) architecture to examine whether sparse expert routing improves model performance and representation capacity.

Through these experiments, we aim to analyze how recent architectural improvements influence the performance of BERT under identical training conditions.

2 Method

This study investigates three architectural modifications to the original BERT architecture: SwiGLU, Rotary Positional Encoding (RoPE), and Mixture-of-Experts (MoE). Each modification was applied independently while keeping all other components unchanged.

2.1 SwiGLU

SwiGLU is a gated activation function that extends the Gated Linear Unit (GLU) by replacing the sigmoid gate with the Swish activation function. Given an input vector (x), SwiGLU is defined as:

$$SwiGLU(x) = Swish(xW_1) \odot (xW_2) \quad (1)$$

where $\odot$ denotes element-wise multiplication and

$$Swish(x) = x\sigma(x). \quad (2)$$

Compared to GELU used in the original BERT, SwiGLU provides a learnable gating mechanism that can improve information flow and representation learning.

2.2 Rotary Positional Encoding (RoPE)

RoPE incorporates positional information directly into the self-attention mechanism by applying rotational transformations to query and key vectors. For a token position (m), the transformed query vector is given by:

$$RoPE(q_m) = R_m q_m, \quad (3)$$

where R_m is a rotation matrix determined by the token position. The attention score becomes

$$Attention(q, k) = (R_m q)^T (R_n k), \quad (4)$$

which naturally encodes relative positional relationships between tokens. Unlike learned positional embeddings, RoPE enables position information to be integrated directly into attention computation.

2.3 Mixture-of-Experts (MoE)

MoE replaces the standard feed-forward network with multiple expert networks and a routing mechanism. Given (N) experts (E_i), the output is computed as

$$y = \sum_{i=1}^N G(x)_i E_i(x), \quad (5)$$

where ($G(x)$) is a gating function that assigns routing weights to each expert. In practice, only a small subset of experts is activated for each token, reducing computational cost while increasing model capacity.

Compared to the standard FFN used in BERT, MoE enables the model to learn more specialized representations through sparse expert routing.

3 Experiments

3.1 Experimental Settings

Unless otherwise specified, all experiments were conducted under the same training configuration.

We used a Transformer-based BERT architecture with 3 encoder layers, 4 attention heads, a hidden dimension of 128, and a feed-forward dimension of 1024. The maximum sequence length was set to 128, and the dropout rate was fixed to 0.1.

For optimization, we used the Adam optimizer with an initial learning rate of 1×10^{-4} . A cosine learning rate scheduler with warm-up was applied throughout training. The batch size was fixed to 32, and all models were trained for 10 epochs.

Pre-training was performed using the Masked Language Modeling (MLM) and Next Sentence Prediction (NSP) objectives. Following the original BERT setting, 15

All ablation studies were conducted under identical conditions while independently modifying specific architectural components. Specifically, we evaluated the effects of replacing GELU with SwiGLU, substituting the standard feed-forward network with Mixture-of-Experts (MoE), and replacing learned positional embeddings with Rotary Positional Encoding (RoPE).

3.2 Dataset

We used the Korean Wikipedia corpus (kowiki.txt) for pre-training. The dataset consists of large-scale Korean text collected from Wikipedia articles and covers a wide range of domains. Following previous language model pre-training studies, the corpus was

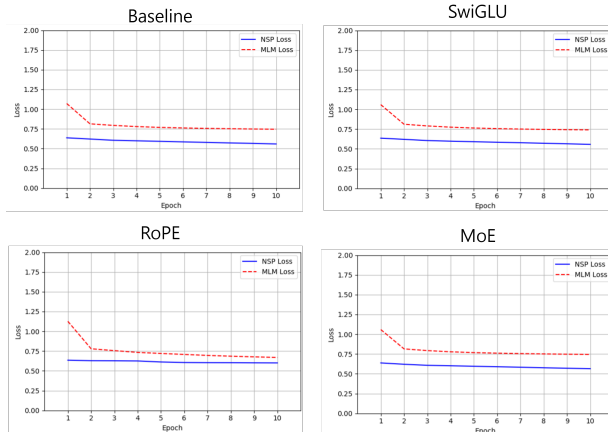


Figure 1: Training loss curves of the baseline BERT model and the three architectural variants (SwiGLU, RoPE, and MoE) during pre-training. The blue solid line represents the NSP loss, while the red dashed line denotes the MLM loss.

used as an unlabeled dataset for the MLM and NSP objectives.

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4 Result

In this section, we analyze the effect of each architectural modification on the performance of BERT. Specifically, we compare the baseline model with variants incorporating SwiGLU, RoPE, and Mixture-of-Experts (MoE). All experiments were conducted using the same training configuration described in Section 2.

The NSP and MLM training losses for the baseline BERT model and the three architectural variants (Figure 1). All models show stable convergence throughout the training process, indicating that the proposed modifications do not negatively affect optimization stability. Compared to the baseline model, SwiGLU and MoE achieve slightly lower MLM losses in the later epochs, suggesting improved learning efficiency and representation learning capability. In contrast, RoPE exhibits a similar convergence pattern but does not provide a noticeable reduction in training loss.

Table 1: Common hyperparameters used in all experiments.

Parameter	Value
Encoder Layers	3
Hidden Dimension	128
Attention Heads	4
FFN Dimension	1024
Maximum Sequence Length	128
Dropout Rate	0.1
Batch Size	32
Learning Rate	1×10^{-4}
Optimizer	Adam
Scheduler	Cosine Decay
Training Epochs	10
MLM Mask Ratio	15%

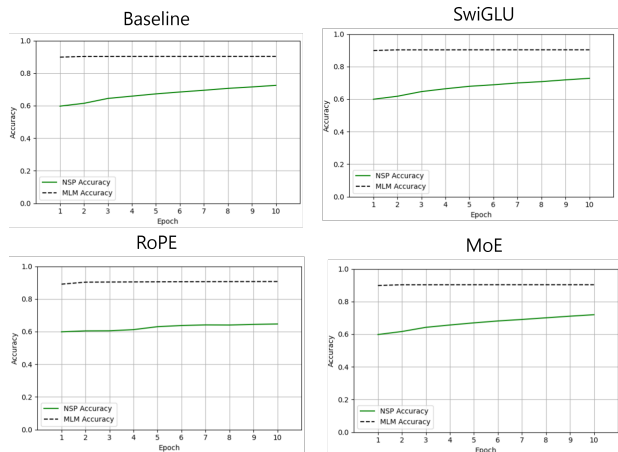


Figure 2: Training Accuracy curves of the baseline BERT model and the three architectural variants (SwiGLU, RoPE, and MoE) during pre-training. The blue solid line represents the NSP loss, while the red dashed line denotes the MLM loss.

Table 2: Experimental configurations used in the ablation study. All settings were kept identical except for the modified component.

Model	Act.	Pos. Enc.	FFN
BERT (Baseline)	GELU	Learned PE	FFN
BERT + SwiGLU	SwiGLU	Learned PE	FFN
BERT + RoPE	GELU	RoPE	FFN
BERT + MoE	GELU	Learned PE	MoE

The NSP accuracy gradually increases for all models as training progresses, while MLM accuracy remains consistently high throughout training (Fig 2). Among the evaluated methods, SwiGLU and MoE achieve slightly higher NSP accuracies than the baseline model, whereas RoPE demonstrates comparable performance. These results indicate that modifications to the activation function and feed-forward network have a greater impact on model learning dynamics than positional encoding in the current experimental setting.

5 Conclusion and Discussion

The results indicate that all architectural variants were successfully trained and exhibited stable convergence during pre-training. Compared to the baseline BERT model, SwiGLU and MoE achieved slightly lower MLM losses and higher NSP accuracies, suggesting improved optimization efficiency and representation learning capability. In contrast, replacing the learned positional embedding with RoPE did not provide a significant performance gain under the current experimental setting.

Although the performance differences were relatively small, the results suggest that modern architectural improvements developed for large-scale language models can also be applied to smaller BERT-based models. Among the evaluated methods, SwiGLU demonstrated the most consistent improvement across both loss and accuracy metrics, while MoE showed comparable performance with the potential benefit of increased model capacity through sparse expert routing.

These findings indicate that activation functions

and feed-forward network designs have a greater impact on BERT pre-training performance than positional encoding methods in small-scale settings.

Although the proposed modifications demonstrated promising results, the experiments were conducted on a relatively small-scale BERT model and a single pre-training dataset. Therefore, additional studies are required to validate the generalizability of these findings. Future work will investigate these architectural modifications under a wider range of experimental settings, including larger model sizes, diverse datasets, and downstream NLP tasks, to better understand their effectiveness and scalability.

6 Reference

Devlin, J., Chang, M. W., Lee, K., Toutanova, K. (2019, June). Bert: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers) (pp. 4171-4186).